

---

# **Repair Manual**

## ***911 Carrera 4*** **(993)**

### **Volume IV: Chassis**

## **IV Chassis**

### **4 Chassis**

|   |                                        |            |
|---|----------------------------------------|------------|
| 4 | Running gear overview of 911 Carrera 4 | .. 4 - 101 |
| 4 | Notes on repair descriptions           | .. 4 - 105 |
| 4 | Checks / Notes on four-wheel drive     | .. 4 - 107 |

### **40 Front Wheel Suspension, Drive Shaft**

|          |                                           |          |
|----------|-------------------------------------------|----------|
| 40 41 19 | Removing and installing front drive shaft | 40 - 101 |
| 40       | Front drive shafts with centering         | 40 - 105 |

### **46 Brake, Mechanical**

|    |                                                       |          |
|----|-------------------------------------------------------|----------|
| 46 | Technical data                                        | 46 - 101 |
| 46 | Technical data Carrera 4S (break disc wear dimension) | 46 - 103 |

## 40 41 19 Removing and installing front drive shaft

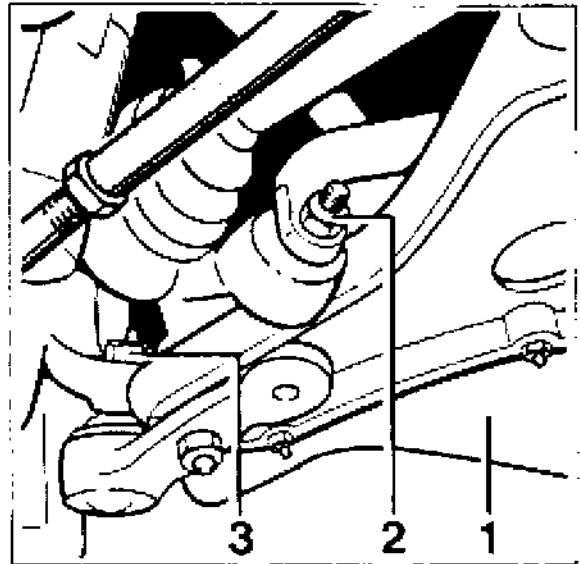
Includes: checking drive shaft runout (constant velocity joint runout) in the case of drive shafts without centering. In the case of drive shafts with centering (see pages 40-105 and 40-106), it is not necessary to check the runout

### Removal

1. Before lifting the vehicle, loosen connection of drive shaft to wheel.  
When doing so, also apply the brakes.
2. Remove front wheel and underside panel.
3. Disconnect brake cooling duct from control arm (1).  
Disconnect stabilizer mount from stabilizer (2).  
Loosen joint carrier (ball joint) from wheel carrier (3). When loosening the fastening nut, hold Torx screwdriver (special tool 9546) **against bolt**.  
Press off ball joint with puller (ball joint extractor) - special tool 9560.

### Note

To prevent damage to rubber sleeve of ball joint, apply tire mounting paste to sleeve and extractor in area concerned. Then insert the extractor from the front.

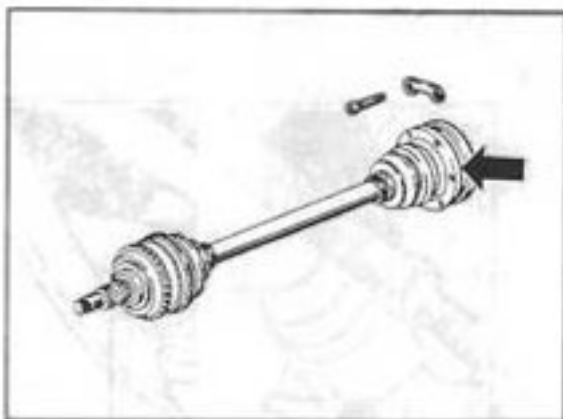


1093-40

4. Unscrew pan head screws of drive shaft from transmission flange.  
**Take note of drawing on following page.**

### Note

The CV joint may fall apart if the dust cover holder (arrow) is removed or damaged.  
The joint must therefore be protected against falling apart by 2 8 mm bolts and nuts during removal, installation and shipment.



1094-40

5. Pull out drive shaft.

#### Note

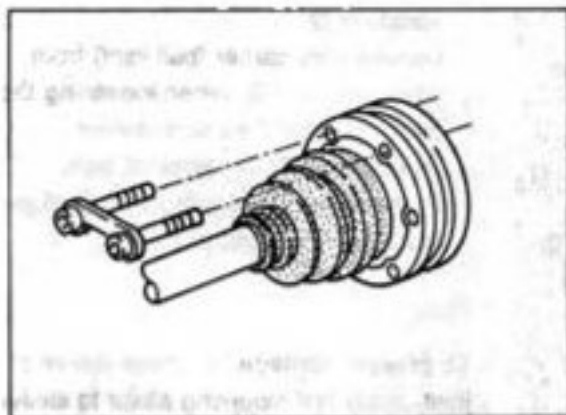
As considerable force is needed to move the drive shaft in the teeth of the wheel hub, loosen the drive shaft using a copper drift. With unfavorable tolerances, it may be necessary to push the drive shaft out of the hub using a wheel hub puller (e.g. a Kiann or Schrem tool)

#### Installation

1. Grease the drive shaft teeth with Optimoly HT.
2. Insert pan head screws with washers in joint flange.  
Insert drive shaft and install pan head screws.

**Caution:** before the pan head screws on the halfshaft flange are finally tightened, the runout of drive shafts without centering must be checked (page 40-103).

In the case of drive shafts with centering (see pages 40-105 and 40-106), checking of the runout (item 3 on page 40-103) is not necessary.



775-42

3. Proceed as follows to check **runout of drive shaft**.

Attach dial gauge with holder  
(e.g. VW 387) to side member.

Apply dial gauge to CV joint between the  
two steel covers, next to the die-stamped  
part no.

Before making the measurement, remove  
any dirt or tectyl residue.

- Measure runout (turn wheel hub and read  
value off gauge).

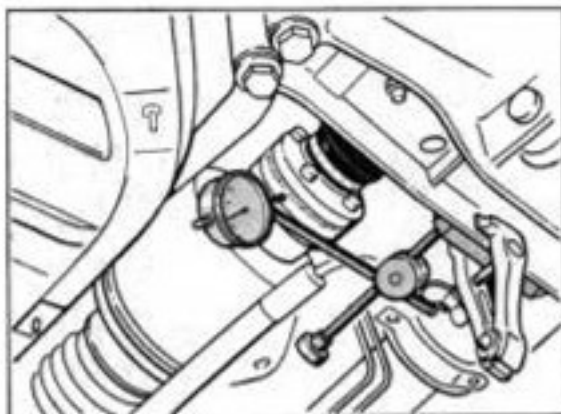
If the runout measured is **more than  
0.2 mm**, loosen the screws slightly and  
center the joint.

When the runout is acceptable (**0.2 mm  
maximum**), tighten the pan head screws  
to 42 Nm, (31 ftlb.).

4. Mount joint carrier (ball joint) on wheel  
carrier. The ball joint cone and the wheel  
carrier must be free from grease.  
Replace fastening nut (new part).  
When tightening fastening nut, hold special  
tool 9546 (Torx screwdriver) against bolt.

5. Assemble brake ventilation duct and stabi-  
lizer mount, using new fastening nuts.

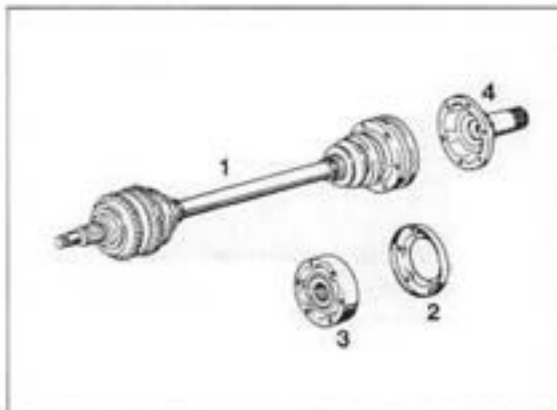
6. Tighten all screws and bolts to specified  
torque.  
Install wheel and underside panel.



## 40 Front drive shafts with centering

Since June 9, 1995, four-wheel-drive vehicles (911 Carrera 4 and 911 Turbo) have been fitted with drive shafts (no. 1) which are centered in the halfshaft flange (no. 4) of the front axle final drive unit by their caps (no. 2).

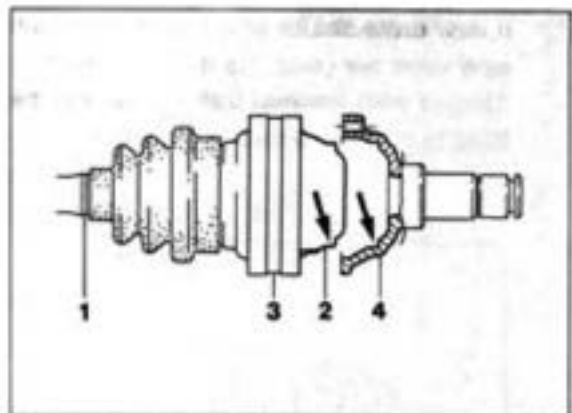
It is therefore no longer necessary to check runout when installing drive shafts. Precise introduction dates (and vehicle ident. nos.) are given in Technical Information, Group 4, No. 3/95.



2274-42

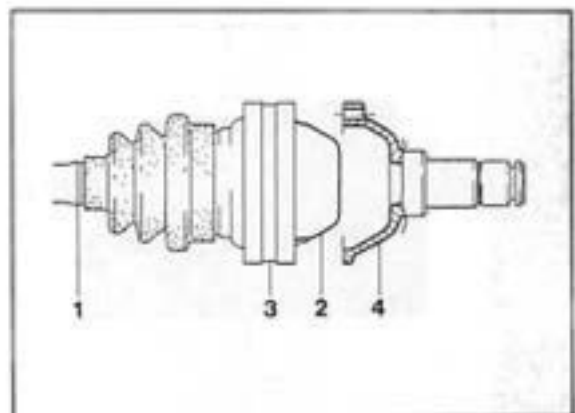
Centering is by means of the cap (2) on the drive shaft (1) and the halfshaft flange (4), which is designed to fit the cap, on the final drive unit.

New (with centering - arrows)



2275-42

Old (without centering)

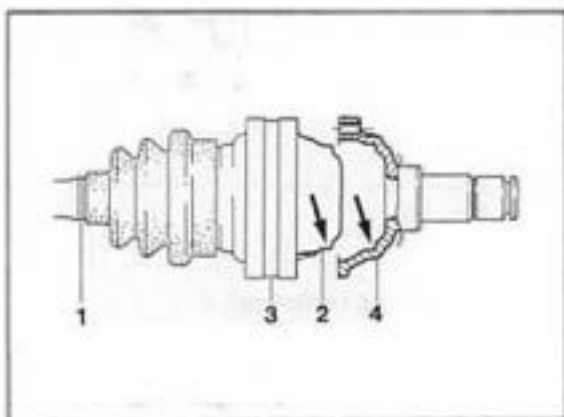


2276-42

For repairs where spare parts are needed and where the customer has complained of vibrations at 180 to 200 kph, only parts (drive shafts or caps and halfshaft flanges) with **centering** (arrows) must be used. With these parts, the runout checks on the drive shafts described on page 40-103 are no longer required.

The **old** drive shafts without centering may be converted to centering by fitting the **new cap**.

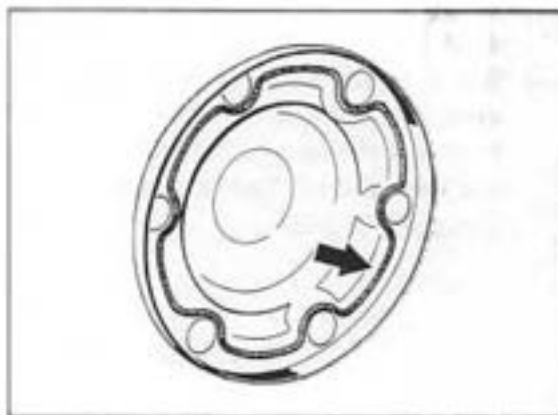
If **new drive shafts** or **old drive shafts with new caps** are used, the **new halfshaft flanges with internal fitting** must also be fitted to the front axle final drive unit.



2275-42

**Caution:** When installing the cap, the sealing surface against the joint must be **sealed** using sealing gel (000 043.203.47) (arrow).

**Important note:** The sealing gel must also be used on the caps of the rear axle drive shafts.



2277-42

**Further information** (allocation / part nos.) is given in Technical Information, Group 4, No. 3 / 95.

## Technical data

| Designation                              |            | Observations, dimensions<br>911 Carrera 4                                                                                                                                                                                                 | Wear limit<br>911 Carrera 4 |
|------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Operating brakes<br/>(foot brake)</b> |            | Hydraulic dual-circuit brake system with front axle/rear axle circuit division. Hydraulic brake booster, internally vented and drilled brake discs with four-piston fixed calipers on front and rear axles. ABS / ABD fitted as standard. |                             |
| Brake booster                            |            | Hydraulic                                                                                                                                                                                                                                 |                             |
| Boost factor                             |            | 4.8                                                                                                                                                                                                                                       |                             |
| Brake master cylinder                    | front dia. | 25.4 mm                                                                                                                                                                                                                                   |                             |
|                                          | rear dia.  | 25.4 mm                                                                                                                                                                                                                                   |                             |
|                                          | stroke     | 17/15 mm                                                                                                                                                                                                                                  |                             |
| Proportioning valve                      |            |                                                                                                                                                                                                                                           |                             |
| Switchover pressure reducing factor      |            | 45 bar - 0.46                                                                                                                                                                                                                             |                             |
| Brake disc diameter                      | front      | 304 mm                                                                                                                                                                                                                                    |                             |
|                                          | rear       | 299 mm                                                                                                                                                                                                                                    |                             |
| Effective brake disc diameter            | front      | 251 mm                                                                                                                                                                                                                                    |                             |
|                                          | rear       | 246 mm                                                                                                                                                                                                                                    |                             |
| Piston diameter in caliper               | front      | 2 x 44 + 2 x 36 mm                                                                                                                                                                                                                        |                             |
|                                          | rear       | 2 x 30 + 2 x 28 mm                                                                                                                                                                                                                        |                             |
| Brake pad area                           | front      | 250 cm <sup>2</sup>                                                                                                                                                                                                                       |                             |
|                                          | rear       | 172 cm <sup>2</sup>                                                                                                                                                                                                                       |                             |
| Total brake pad area                     |            | 422 cm <sup>2</sup>                                                                                                                                                                                                                       |                             |
| Pad thickness                            | front      | approx. 11.0 mm                                                                                                                                                                                                                           | 2 mm                        |
|                                          | rear       | approx. 12.0 mm                                                                                                                                                                                                                           | 2 mm                        |



| Designation                                                        | Observations, dimensions<br>911 Carrera 4           | Wear limit<br>911 Carrera 4 |
|--------------------------------------------------------------------|-----------------------------------------------------|-----------------------------|
| Thickness of new brake disc<br>front<br>rear                       | 32 mm<br>24 mm                                      |                             |
| Minimum brake disc thickness<br>* after machining<br>front<br>rear | 30.6 mm<br>22.6 mm                                  | 30.0 mm<br>22.0 mm          |
| Thickness tolerance of brake disc, max.                            | 0.02 mm<br>(new condition 0.01 mm)                  |                             |
| Lateral runout of brake disc, max.                                 | 0.05 mm                                             |                             |
| Lateral runout of wheel hub, max.                                  | 0.04 mm                                             |                             |
| Lateral runout of brake disc when<br>fitted, max.                  | 0.09 mm                                             |                             |
| Surface roughness of brake disc<br>after machining, max.           | 0.006 mm                                            |                             |
| Pushrod play<br>(measured at brake pedal plate)                    | approx. 8 mm                                        |                             |
| <b>Parking brakes (handbrake)</b>                                  | Drum brake, acting mechanically on both rear wheels |                             |
| Parking brake drum diameter                                        | 180 mm                                              | 181 mm                      |
| Brake shoe width                                                   | 25 mm                                               |                             |
| Brake lining thickness                                             | 4.5 mm                                              | 2 mm                        |

- \* The brake disc must only be machined symmetrically, i.e. by an identical amount on both sides.

## 46 Technical data Carrera 4S (Turbo-Look)

| Designation                              |            | Observations, dimensions                                                                                                                                                                                                                   | Wear limit |
|------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>Operating brakes<br/>(foot brake)</b> |            | Hydraulic dual-circuit brake system with front axle/rear axle circuit division. Hydraulic brake booster, internally vented and drilled brake discs with four-piston fixed calipers on front and rear axles. ABS / ABD* fitted as standard. |            |
| Brake booster                            |            | Hydraulic                                                                                                                                                                                                                                  |            |
| Boost factor                             |            | 4.8                                                                                                                                                                                                                                        |            |
| Brake master cylinder                    | front dia. | 25.4 mm                                                                                                                                                                                                                                    |            |
|                                          | rear dia.  | 25.4 mm                                                                                                                                                                                                                                    |            |
|                                          | stroke     | 17/15 mm                                                                                                                                                                                                                                   |            |
| Proportioning valve (2 units)            |            |                                                                                                                                                                                                                                            |            |
| Switchover pressure reducing factor      |            | 40 bar - 0.46                                                                                                                                                                                                                              |            |
| Brake disc diameter                      | front      | 322 mm                                                                                                                                                                                                                                     |            |
|                                          | rear       | 322 mm                                                                                                                                                                                                                                     |            |
| Effective brake disc diameter            | front      | 259.6 mm                                                                                                                                                                                                                                   |            |
|                                          | rear       | 268.4 mm                                                                                                                                                                                                                                   |            |
| Piston diameter in caliper               | front      | 2 x 44 + 2 x 36 mm                                                                                                                                                                                                                         |            |
|                                          | rear       | 2 x 30 + 2 x 28 mm                                                                                                                                                                                                                         |            |
| Brake pad area                           | front      | 302 cm <sup>2</sup>                                                                                                                                                                                                                        |            |
|                                          | rear       | 250 cm <sup>2</sup>                                                                                                                                                                                                                        |            |
| Total brake pad area                     |            | 552 cm <sup>2</sup>                                                                                                                                                                                                                        |            |
| Pad thickness                            | front      | approx. 11.0 mm                                                                                                                                                                                                                            | 2 mm       |
|                                          | rear       | approx. 11.0 mm                                                                                                                                                                                                                            | 2 mm       |

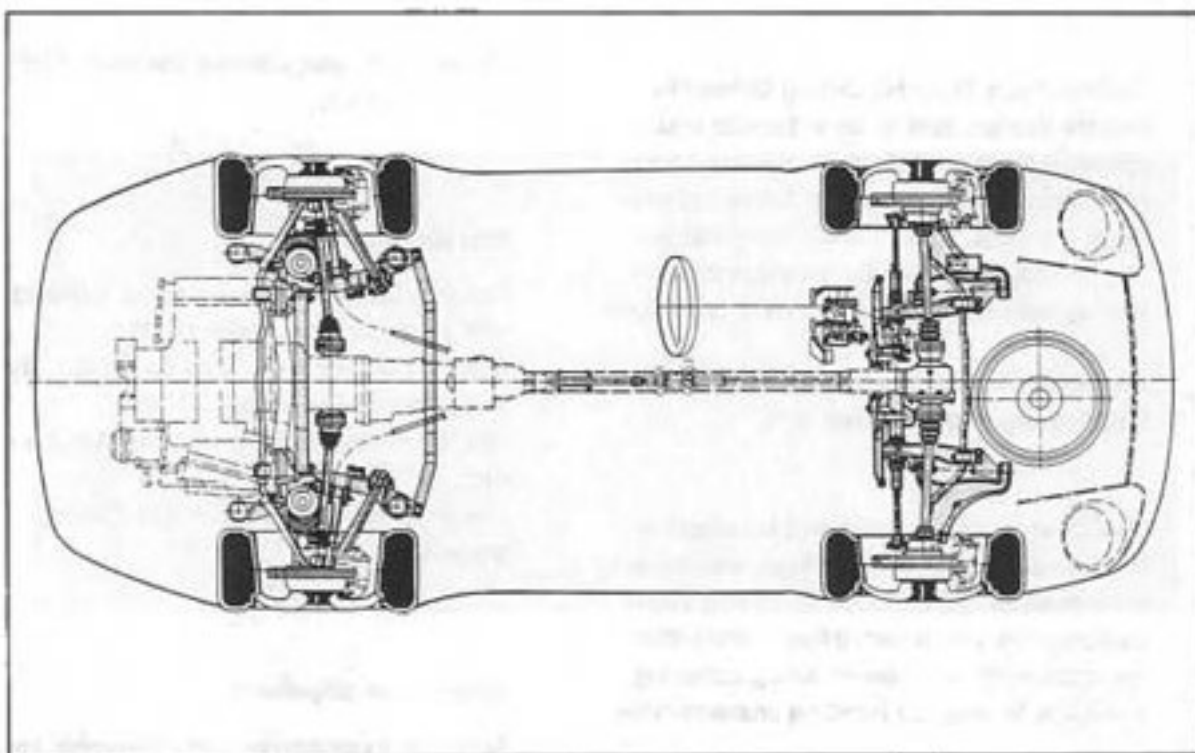
\* ABD = automatic brake differential

| Designation                                              | Observations, dimensions                            | Wear limit |
|----------------------------------------------------------|-----------------------------------------------------|------------|
| Thickness of new brake disc                              |                                                     |            |
| front                                                    | 32 mm                                               |            |
| rear                                                     | 28 mm                                               |            |
| Minimum brake disc thickness<br>* after machining        |                                                     |            |
| front                                                    | 30.6 mm                                             | 30.0 mm    |
| rear                                                     | 26.6 mm                                             | 26.0 mm    |
| Thickness tolerance of brake disc, max.                  | 0.02 mm<br>(new condition 0.01 mm)                  |            |
| Lateral runout of brake disc, max.                       | 0.05 mm                                             |            |
| Lateral runout of wheel hub, max.                        | 0.04 mm                                             |            |
| Lateral runout of brake disc when<br>fitted, max.        | 0.09 mm                                             |            |
| Surface roughness of brake disc<br>after machining, max. | 0.006 mm                                            |            |
| Pushrod play<br>(measured at brake pedal plate)          | approx. 8 mm                                        |            |
| <b>Parking brakes (handbrake)</b>                        | Drum brake, acting mechanically on both rear wheels |            |
| Parking brake drum diameter                              | 180 mm                                              | 181 mm     |
| Brake shoe width                                         | 25 mm                                               |            |
| Brake lining thickness                                   | 4.5 mm                                              | 2 mm       |

The brake disc must only be machined symmetrically, i.e. by an identical amount on both sides.

#### 4 Running gear overview of 911 Carrera 4

Just like the 911 Carrera (rear-wheel drive 993), the 911 Carrera 4 (four wheel drive 993) is offered with a variety of running gear versions (standard running gear / sports-type running gear M 030 for Coupé only / lowered running gear M 033). Components and running gear tuning of the 911 Carrera (993) and 911 Carrera 4 (993) differ only slightly from each other.



##### Four-wheel drive (General)

The 911 Carrera 4 is fitted with a permanent four-wheel drive system with **variable** power distribution to the front and rear wheels.

Power is distributed to the wheels across a viscous multi-disc clutch to reflect the wheel speed difference of the front and rear wheels. This ensures that the front wheels are given only enough drive torque to ensure optimum propulsion even under adverse road conditions.

The front wheels are driven by the viscous multi-disc clutch and a central shaft to the front-axle final drive.

The viscous multi-disc clutch is housed in the front transmission housing of the six-speed manual transmission. The design of the six-speed manual transmission is based on the 911 Carrera (993) version.

Additionally, a Dynamic Driving Differential System that consists of an automatic brake operation feature (ABD) and a load-dependent (mechanical) transverse lock differential provides improved traction and driving stability. The limited-slip differential provides different locking values for acceleration and under load:

**Locking value during acceleration 25%**  
**Locking value under load 40%**

The lower locking value during acceleration considerably improves cornering characteristics. An increased locking value for driving under load reduces yaw (oversteering effect) when the accelerator is released during cornering and helps to stabilize handling characteristics.

#### Note

For a detailed description, please refer to Service Information Technik '95. Order No. WKD 499 520 (2 = English).

#### Front axle

The design of the front axle is identical to that of the 911 Carrera.

To provide sufficient space for the drive shafts, the stabilizer mount is now shaped as an angled drop forging.

#### Rear axle

The rear axle was adopted unchanged from the 911 Carrera.

#### Wheels, tires

The 911 Carrera 4 is supplied as standard with 16-inch or (optionally, M No.) with 17-inch wheels in "Cup 93" design (availability as per September, 1994).

The "Carrera 4" script is embossed in the hub-caps.

Tire pressures: same as on 911 Carrera (page 44 - 1).

#### Suspension alignment

Adjustment procedures and adjustment specifications of front and rear axles are the same as on the 911 Carrera (as of page 44 - 3).

#### Brakes - General information

The 911 Carrera 4 is fitted with a dual-circuit brake system (front/rear axle division) **with hydraulic brake booster**. ABS and the dynamic lock system (**consisting of ABD and limited-slip differential**) are supplied as standard.

#### Front brakes

The front brakes were adopted from the 911 Carrera (drilled brake disc, four-piston light-alloy brake calipers).

The brake pad grade is the same as for the 911 Carrera (modified as of MY '95 / please refer to spare parts catalog). Both sides of the brake pads are monitored for wear by brake pad wear indicators.

The brake caliper is painted in a **titanium color**.

### Rear wheel brakes

The rear brakes are fitted with the drilled brake disc adopted from the 911 Carrera. The four-piston light-alloy caliper is fitted with pistons with a diameter of 28 mm and 30 mm (911 Carrera dias. are 30 mm and 34 mm, respectively).

The brake pad grade is the same as on the 911 Carrera (modified as of MY '95 / please refer to spare parts catalog). Both sides of the brake pads are monitored for wear by brake pad wear indicators.

The brake caliper is painted in a **titanium color**.

### Proportioning valve

Two proportioning valves (for left and right-hand side brakes) are fitted to reduce the braking pressure at the rear axle and to adapt the braking pressure to the wheel load distribution.

Switchover pressure: 45 bar

Reducing factor 0.46

(Identification: 5 ↓ 45).

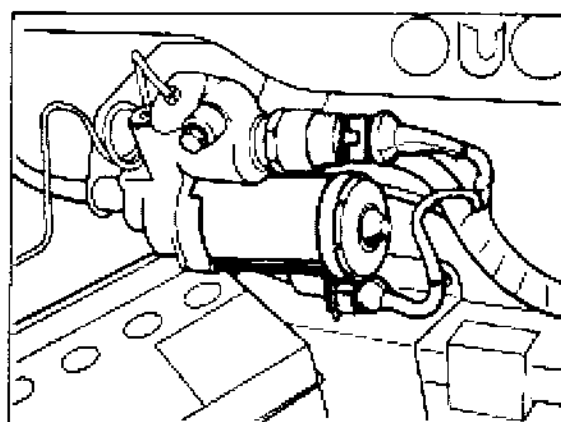
### Brake boost / hydraulic pump

The electro-hydraulic brake booster system known from the 911 Carrera 4 (Model Year '89) is used to provide a brake boost effect. To optimize the system, a modified **hydraulic pump** is fitted.

This pump may also be retrofitted to the 911 Carrera 4 as of Model Year '89.

### Note

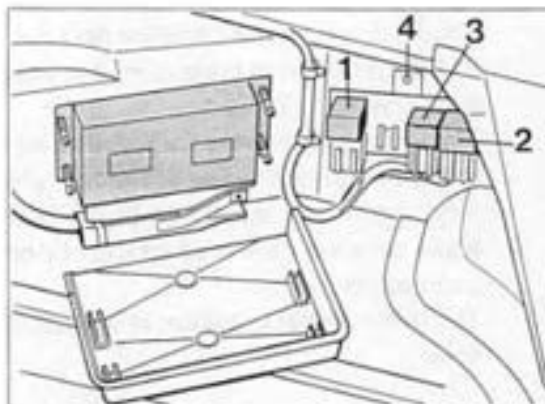
The hydraulic pump relay is no longer located in the Central Electrical System (as in the case of the 911 Carrera 4 as of Model Year '89) but rather in the right-hand front end of the luggage compartment (refer to the below paragraph on relays).



2000-47

### ABS / ABD hydraulic unit

The 911 Carrera 4 is fitted with the optimized ABS/ABD hydraulic unit that had already been fitted to 911 Carrera vehicles with ABS/ABD system in March 1994 (refer to Technical Information 3/94 of May 27, 1994). The ABS/ABD control unit (993.618.127.00) was adapted to the modified hydraulic unit and the four-wheel drive system. This control unit is also used on the 911 Carrera (rear-wheel drive) with ABD - as of MY '95. To help identification, a sticker with a **red outline** is attached to the control unit.



1773C-47

### Relays

Three relays are mounted on a bracket in the right-hand front end of the luggage compartment to control the return pump and the hydraulic pump and to supply power to the solenoids in the hydraulic unit.

- 1 = Return pump relay
- 2 = Solenoid relay
- 3 = Hydraulic pump relay
- 4 = Body-bound rivet

### Note

To remove the cover from the relay bracket, push out the pin in the body-bound rivet and remove the complete bracket assembly. The pin may be retrieved from behind the bracket plate by applying compressed air.

### Steering system

The steering system was retained from the 911 Carrera without any changes.

## **4 Notes on repair descriptions**

### **General information**

The 911 Carrera (rear-wheel drive 993) forms the basis of the description of repair, assembly and adjustment operations.

This means that the "911 Carrera 4" Repair Manual (filed after the separation sheet) covers **only operations that affect the 911 Carrera 4**.

### **Notes on tightening torques**

Most tightening torques of the 911 Carrera and 911 Carrera 4 are identical.

Differing or additional tightening torques are included in the tables in the corresponding repair groups of the 911 Carrera (rear-wheel drive 993).

### **Notes on suspension alignment (Repair Group 44)**

The adjustment operations and specifications for front and rear axles are the same as for the 911 Carrera (rear-wheel drive 993).

### **Notes on brake boosting (Repair Group 47)**

The hydraulic brake booster system is also described in the manuals on the 911 Carrera (rear-wheel drive 993).



## 4 Checks / Notes on four-wheel drive

### Balancing the wheels on the vehicle

All four wheels must be off the ground and able to spin freely when the wheels are precision balanced on the vehicle.

### Performance tests

Performance tests must only be run on 4-roller dynamometers with rpm coupling. When running tests on 2-roller dynamometers, interrupt the connection between front and rear axle at the central shaft.

### Brake tests

Brake tests must only be run on roller dynamometers or plate dynamometers.

If no four-wheel drive brake dynamometer (roller dynamometer) is available, make sure the below limits are not exceeded on conventional roller dynamometers:

|               |            |
|---------------|------------|
| Test speed    | 5 mph      |
| Test duration | 20 seconds |

### Towing

If the vehicle has to be towed with the front or rear axle off the ground, the wheels of the raised axle must be able to rotate freely.